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Hierarchical Semantic Concept Modeling for Generalizable Myocardial Pathology Segmentation on Multisequence CMR Images

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***Abstract*—Myocardial pathology segmentation (MyoPS) aims to accurately quantify myocardial scar and edema, which is critical for precise assessment of myocardial infarction (MI) severity. Recent deep learning (DL) methods have shown promising performance. However, due to cross-domain distribution shifts, many methods exhibit limited generalizability because it is challenging to learn domain-invariant pathology information, resulting in degraded performance on unseen data. This work proposes a hierarchical semantic concept segmentation framework, referred to as HSCM-Net. This framework uses concept modeling of multisequence cardiac magnetic resonance (CMR) images and labels to integrate complementary information and enhance the generalizability of MyoPS. Specifically, the CMR images are first decomposed into three independent concept variables, and hierarchical priors are assigned to explicitly force them to model the sequence-irrelevant concept of shape, the semantic concept of pathology, and the domain-specific concept of appearance. Then, the anatomy segmentation and pathology segmentation are modeled as global and local concepts only related to shape and pathology, respectively. Finally, a variational inference (VI) framework is developed to approximate the posteriors of these conceptual variables, and the inference process is implemented with a deep neural network. Results on three-domain multisequence CMR datasets proved the advantageous generalizability of our HSCM-Net, which achieves a MyoPS Dice score of 0.577 on unseen target domains. Our code is released at https://github.com/Dongjinweicn/HSCMNet**

***Index Terms*—Generalizability, hierarchical concept, myocardial pathology segmentation (MyoPS), semantic modeling, variational inference (VI).**

I. Introduction

ACUTE myocardial infarction (MI) is a prevalent and highly detrimental cardiovascular disease with persistently high mortality rates worldwide, causing millions of deaths annually [1]. It induces irreversible damage to the myocardium (MYO) due to acute ischemia [2]. The utilization of CMR imaging, renowned for its capacity to deliver high-resolution, multisequence data, emerged as a fundamental tool for evaluating myocardial pathology [3]. For example, balanced steady-state free precession (bSSFP) cine sequences provide distinct anatomy contours, while late gadolinium enhancement (LGE) and T2-weighted (T2) CMR images highlight scar and edema areas, respectively [4]. Precisely identifying the damaged areas in the MYO contributes to accurate diagnosis, timely treatment, and effective prognostic management of MI [5]. However, due to the complex shape, variable location, and blurry boundary of pathology regions, manual delineation remains an exceedingly time-consuming and labor-intensive task. Consequently, there is a growing focus on automatic myocardial pathology segmentation (MyoPS) approaches in research endeavors.

Thanks to the advancements in deep learning (DL), many studies employed convolutional neural network (CNN) for the automatic segmentation of myocardial pathology [11], [12]. Several works utilized single-sequence (e.g., LGE or T2) CMR for segmenting a type of pathology, specifically for scar or edema [13], [14], [15]. Beyond single-sequence CMR images, multisequence CMR images provide richer pathology information. The effective fusion of information from multiple sequences for simultaneous segmentation of scar and edema holds practical clinical significance [16]. However, current multimodality fusion methods exhibit limited robustness and struggle to effectively capture the complex interplay between diverse modalities. Furthermore, recent research focused on a two-stage segmentation strategy, emphasizing the spatial relationship between pathology and MYO [17]. The first stage involves segmenting the left ventricle (LV) MYO, followed by the

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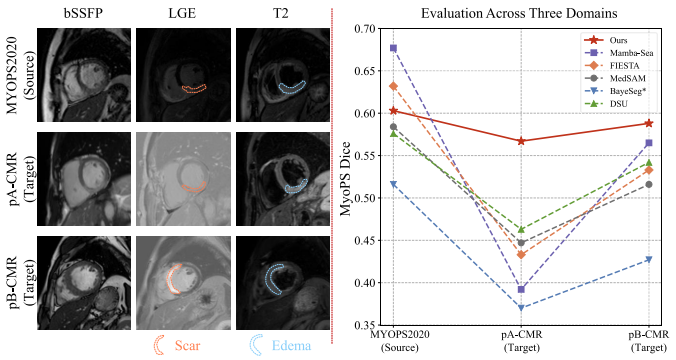


Fig. 1. Typical images from three-domain multisequence CMR datasets, and a comparison of MyoPS performance of our method against Mamba-Sea [6], FIESTA [7], MedSAM [8], BayeSeg⋆ [9], and DSU [10].

segmentation of scar and edema on the MYO. However, the success of these methods relies on the assumption that the training data (i.e., source) and the test data (i.e., target) share the same distribution, but they could fail to generalize on unseen data with domain shifts.

To solve such an issue, many methods are proposed for domain generalization (DG). The existing ones can be divided into three groups, i.e., data level, learning level, and representation level [18]. One group, data level, enriches the distribution of training by augmenting or generating new samples from the source data [19], [20], [21]. However, these methods perform poorly in the presence of significant domain gaps [22]. Another group, learning level, focuses on optimizing model training to learn general knowledge from training data, mainly involving ensemble learning and meta-learning [23], [24], [25]. However, the generalizability of these methods may be limited by the quantity of source domains [26]. The third group, representation level, aims to learn domain-invariant features by aligning or disentangling features [27], [28], [29]. Current DG approaches based on extracting domain-invariant representations show promising performance in cross-domain medical organ segmentation tasks. However, as shown in Fig. 1, due to the variability in size, shape, and morphology of myocardial pathology regions, learning domain-invariant pathology information is extremely challenging, which leads to limited generalizability of many existing DG methods for pathology segmentation on unseen domains.

To overcome the challenges mentioned above, this work proposes a generalizable Bayesian DL method based on a hierarchical semantic concept model for MyoPS, referred to as HSCM-Net. Concretely, the information from multisequence CMR images is first fused and represented as three concept variables. The first one is sequence-irrelevant, representing the semantic concept of anatomy shape information; the second is domain-invariant, representing a higher level semantic concept of myocardial pathology information; and the third is domain-specific for representing the concept of appearance information. Then, as the three concepts are unconstrained, hierarchical priors are assigned to model their statistics. Afterward, due to the intractability of directly computing the concept posteriors, a variational inference (VI) is employed to approximate them. Finally, a DL framework is designed to implement the variational method and infer the posterior distributions of these concepts from given multisequence CMR images. In addition, based on the positional relationship among MYO, scar, and edema, a MYO-pathology constraint loss and a scar-edema constraint loss are introduced. These constraint losses, combined with the variational loss and cross-entropy loss, are jointly used to train the deep inference neural network.

The main contributions of our work are summarized as follows.

1) A novel generalizable MyoPS framework, called HSCM-Net, is proposed through hierarchical semantic concept modeling of multisequence CMR images and labels.
2) The framework integrates information from multisequence CMR images by modeling shape, pathology, and appearance concepts, facilitating the comprehensive extraction of pathology information.
3) The generalizability of HSCM-Net is validated on three-domain multisequence CMR datasets, and its effectiveness is further confirmed through ablation analysis of concept modeling.

The remainder of this article is organized as follows. Section II reviews related work; Section III introduces our hierarchical semantic concept model; Section IV presents experimental results and analysis; and Section V concludes this article, discussing its limitations and potential future directions.

II. Related Works

A. Myocardial Pathology Segmentation

Recently, DL was extensively employed in the field of medical image segmentation, yielding satisfactory performance [30]. An increasing number of studies utilized CNN to achieve automated segmentation of myocardial pathology from CMR images [31], [32], [33]. For example, Fahmy et al. [14] adapted a deep 3-D U-Net architecture for the quantification of scar from LGE images. Furthermore, due to the positional relationship between pathology and LV MYO, Zabihollahy et al. [34] employed a semiautomatic CNN method based on myocardial boundary delineation to segment scar from LGE images. Moreover, in addition to utilizing single-sequence CMR image for scar segmentation, leveraging the abundant pathology information provided by multisequence CMR images can simultaneously segment scar and edema. Recent researchers have focused on employing a two-stage segmentation strategy, following a coarse-to-fine approach by first segmenting the MYO and then localizing scar and edema. Zhang et al. [35] designed a two-stage framework composed of a myocardial segmentation network and pathology segmentation network. Wang et al. [36] incorporated reinforcement learning into the fine segmentation of a two-stage process to perform weighted deep supervision. Ding et al. [37] proposed a combined computing method, first fusing multisequence information through registration for myocardial segmentation, and then utilizing U-Net with myocardial constraints to segment scar

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and edema. Qiu et al. [38] first designed a prior consistency module to segment the MYO, and then designed a feature fusion module to integrate information from five sequences for MyoPS. However, while these methods perform satisfactorily on a particular dataset, they are trained and tested on data from a single domain, thus limiting their generalizability.

B. DG for Medical Image Segmentation

To address the out-of-distribution problem in medical image analysis, DG methods attempt to learn from one or multiple source domains to generalize well on unseen target domains. In general, these DG methods can be broadly classified into three groups, i.e., data-level, learning-level, and representation-level approaches [18]. *For data-level approaches* [6], Taleb et al. [39] employed a multimodal puzzle to confuse medical image modalities, where patches from different modalities are randomly assembled. Zhou et al. [40] conducted random amplitude mixup in a frequency space to synthesize new images. Ouyang et al. [41] proposed a causality-inspired data augmentation method to simulate the data acquisition process of various medical images. Su et al. [42] introduced a saliency-balancing location-scale augmentation algorithm for both global and local data augmentation. *For learning-level approaches*, Liu et al. [43] presented a shape-aware meta-learning approach, incorporating compactness and smoothness losses to improve the accuracy of the segmentation boundary. Liu et al. [44] proposed a semi-supervised meta-learning method that explicitly models and disentangles representations to more accurately capture true domain shifts. Liu et al. [45] presented an episodic learning strategy in frequency spaces, improving generalizability through domain information exchange and explicit regularized-guided meta-learning. Hu et al. [25] introduced meta-learning on anatomy-consistent pseudo-modalities, enhancing generalizability by learning vascular structure features. *For representation-level approaches*, feature alignment aims to learn domain-invariant features by minimizing the discrepancy between different domain features. Researchers adapted adversarial learning [46], [47], maximum mean discrepancy (MMD) metric [48], or Kullback–Leibler (KL) divergence [49] to align features across different domains. Feature disentanglement aims to separate extracted representations into domain-specific features and domain-agnostic features. Gu et al. [50] decomposed a feature into domain-invariant anatomy features and domain-specific style codes, achieving disentanglement of style codes through a contrastive loss. Similarly, Bi et al. [51] proposed MI-SegNet that extracts anatomy and domain features from images and minimizes the mutual information between these features. Although these representation-level methods achieved promising results in various cross-domain medical organ segmentation tasks, there are relatively few DG methods for cross-domain MyoPS.

C. Bayesian Statistical Modeling

Bayesian statistical modeling applies the principles of Bayesian inference to data analysis, where parameters of statistical models are updated based on prior knowledge and observed data [52]. The integration of Bayesian modeling with DL has gained increasing attention. The variational autoencoder (VAE), proposed by Kingma and Welling [53], is a notable example. Its underlying statistical model is a simple Bayesian latent variable model, mapping high-dimensional observed values through a CNN to low-dimensional latent variables following a normal distribution. Choi et al. [54] further incorporated VAE into an active learning framework, leveraging Bayesian rules to model the deep feature representation of classifiers based on unlabeled data. Gao and Zhuang [55] proposed a Bayesian super-resolution framework where image statistics are modeled as smoothness components, sparse residuals, and noise; then a CNN is utilized to infer their posterior distributions. Furthermore, Gao et al. [9] introduced a Bayesian medical segmentation method (BayeSeg), decomposing images into appearance and shape variables. The organ structures are segmented from the shape through joint statistical modeling, and variational methods for posterior inference are implemented by CNNs. Inspired by Bayesian statistical modeling [9], [55], this study proposes a hierarchical semantic concept modeling approach for MyoPS. Since the variability of pathology size and shape renders the anatomical priors used in BayeSeg unsuitable for myocardial pathology, a concept variable is designed to explicitly model the semantic information of the pathology, forming a distribution of concepts while also representing domain-invariant information.

TABLE I
SUMMARY OF MATHEMATICAL NOTIONS AND CORRESPONDING CONCEPTS

Notation	Concept
$\boldsymbol{y} \in \mathbb{R}^{d_y \times C}$	Multi-sequence CMR images
$\boldsymbol{u} \in \mathbb{R}^{d_y \times K}$	Segmentation label of anatomy
$\boldsymbol{v} \in \mathbb{R}^{d_y \times N}$	Segmentation label of pathology
$\boldsymbol{s} \in \mathbb{R}^{d_y \times 1}$	Shape
$\boldsymbol{o} \in \mathbb{R}^{d_y \times 1}$	Pathology
$\boldsymbol{a} \in \mathbb{R}^{d_y \times C}$	Appearance
$\boldsymbol{z} \in \mathbb{R}^{d_y \times K}$	Segmentation of anatomy
$\boldsymbol{x} \in \mathbb{R}^{d_y \times N}$	Segmentation of pathology
$\boldsymbol{\upsilon} \in \mathbb{R}^{d_y \times 1}$	Shape boundary
$\boldsymbol{\varepsilon} \in \mathbb{R}^{d_y \times 1}$	Pathology boundary
$\boldsymbol{m} \in \mathbb{R}^{d_y \times C}$	Appearance mean
$\boldsymbol{\rho} \in \mathbb{R}^{d_y \times C}$	Appearance inverse variance
$\boldsymbol{\omega} \in \mathbb{R}^{d_y \times K}$	Anatomy segmentation boundary
$\boldsymbol{\eta} \in \mathbb{R}^{d_y \times N}$	Pathology segmentation boundary
$\boldsymbol{\pi} \in \mathbb{R}^{K}$	Anatomical label portion
$\boldsymbol{\tau} \in \mathbb{R}^{N}$	Pathological label portion
$\mathcal{N}(\cdot,\cdot), \mathcal{G}(\cdot,\cdot), \mathcal{B}(\cdot,\cdot)$	Gaussian, Gamma, Beta distributions
$\boldsymbol{\mu}_{\cdot}^{(0)}, \boldsymbol{\sigma}_{\cdot}^{(0)}, \boldsymbol{\phi}_{\cdot}^{(0)}, \boldsymbol{\gamma}_{\cdot}^{(0)}, \boldsymbol{\alpha}_{\cdot}^{(0)}, \boldsymbol{\beta}_{\cdot}^{(0)}$	Hyperparameters of distributions
$\hat{\boldsymbol{\mu}}_{\cdot}/\hat{\boldsymbol{\sigma}}_{\cdot}$	Mean/Deviation of variational distributions

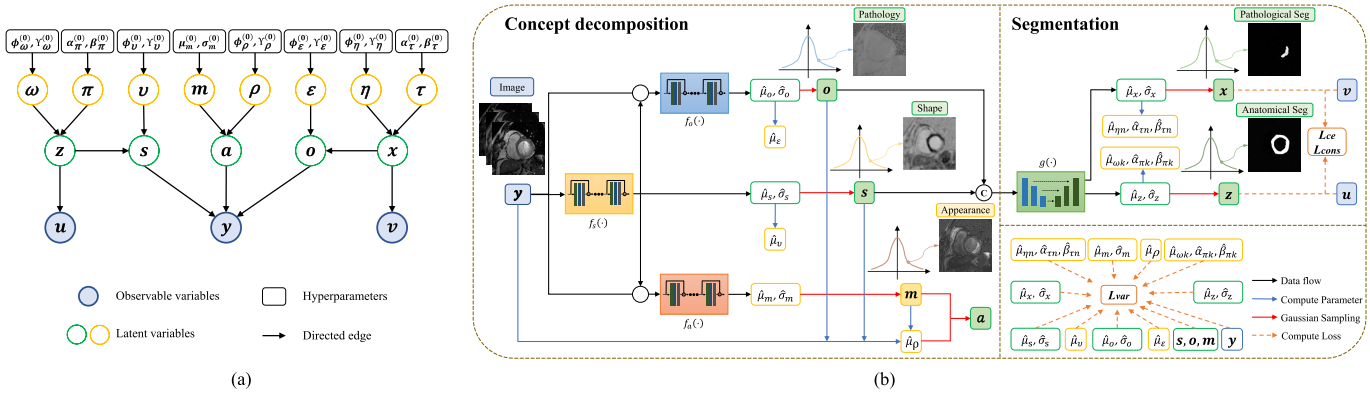
III. METHOD

A. Overview of HSCM-Net

Fig. 2 shows the framework of the proposed HSCM-Net that includes semantic concept modeling and deep inference, and Table I summarizes the notations and concepts utilized in this study. For the semantic concept modeling, a probabilistic graphical model (PGM) is built, as illustrated in Fig. 2(a). Specifically, the multisequence CMR images $\boldsymbol{y}$, concatenated from multiple sequences of CMR images, are first decomposed

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(a) (b)

Fig. 2. Framework of hierarchical semantic concept modeling (HSCM-Net) for MyoPS. (a) PGM of HSCM-Net. Here, the observed image $\boldsymbol{y}$ is decomposed into three latent concepts: shape $\boldsymbol{s}$, pathology $\boldsymbol{o}$, and appearance $\boldsymbol{a}$. The appearance $\boldsymbol{a}$ is governed by its mean $\boldsymbol{m}$ and inverse variance $\boldsymbol{\rho}$; $\boldsymbol{s}$ and $\boldsymbol{o}$ depend on their respective boundaries $\boldsymbol{\upsilon}$ and $\boldsymbol{\varepsilon}$. Anatomy segmentation $\boldsymbol{z}$ and pathology segmentation $\boldsymbol{x}$ are conditioned on their respective boundaries $(\boldsymbol{\omega}, \boldsymbol{\eta})$ and label portions $(\boldsymbol{\pi}, \boldsymbol{\tau})$. (b) DL framework for HSCM-Net. First, given multisequence CMR images, a CNN is employed to infer the posterior distribution of semantic concept shape $(\boldsymbol{s})$. Next, two CNNs are utilized to separately infer the posterior distributions of pathology concept $(\boldsymbol{o})$ and appearance concept mean $(\boldsymbol{m})$ from the residual $\boldsymbol{y} - \boldsymbol{s}$. Then, corresponding random samples are obtained from these distributions. Finally, a U-Net is used to infer the posterior distributions of anatomy segmentation $(\boldsymbol{z})$ and pathology segmentation $(\boldsymbol{x})$ from the concatenation of pathology and shape concept variables, where the obtained random samples serve as predictions during the training process. For detailed information on the network architecture, please refer to Section III-E.

into three concept variables. The first one is the sequence-agnostic global semantic concept, shape $\boldsymbol{s}$, which models the anatomical structures information; the second is the domain-invariant local higher level semantic concept, pathology $\boldsymbol{o}$, which models the scar and edema information; and the third is the domain-specific concept, appearance $\boldsymbol{a}$, which models the style and texture information of the CMR images. Since these concepts represent domain-general knowledge, the model predicting segmentation from these concept variables could improve generalizability. To hierarchically model the statistics of multisequence CMR images and labels, the shape $\boldsymbol{s}$ depends on its boundary $\boldsymbol{\upsilon}$ for detecting the contours of cardiac structure and the anatomy segmentation $\boldsymbol{z}$. The pathology $\boldsymbol{o}$ depends on its boundary $\boldsymbol{\varepsilon}$ for detecting the edges of pathological structure and the pathology segmentation $\boldsymbol{x}$. Meanwhile, the appearance $\boldsymbol{a}$ is modeled to be a Gaussian variable governed by the domain-specific mean $\boldsymbol{m}$ and the domain-specific inverse variance $\boldsymbol{\rho}$. Moreover, the segmentation $\boldsymbol{z}$ is constrained to be globally smoothness and conditioned on its boundary $\boldsymbol{\omega}$ and the label portion $\boldsymbol{\pi}$ of three anatomical classes, the segmentation $\boldsymbol{x}$ is constrained to be locally smoothness and conditioned on its boundary $\boldsymbol{\eta}$ and the label portion $\boldsymbol{\tau}$ of two pathological classes. In addition, the segmentation labels $\boldsymbol{u}$ and $\boldsymbol{v}$ follow the Bernoulli distribution, which are modeled as variables depending on the segmentation $\boldsymbol{z}$ and $\boldsymbol{x}$. Finally, Gamma priors are assigned to $\boldsymbol{\rho}$, $\boldsymbol{\upsilon}$, $\boldsymbol{\omega}$, $\boldsymbol{\eta}$, and $\boldsymbol{\varepsilon}$, a Gaussian prior to $\boldsymbol{m}$, and Beta priors to $\boldsymbol{\pi}$ and $\boldsymbol{\tau}$. For the deep inference, a deep neural network is designed, as illustrated in Fig. 2(b). First, three separate CNNs are utilized to infer the posterior distributions of the shape, pathology, and appearance concepts. Second, another CNN is used to estimate the posterior distribution of anatomy and pathology segmentations, based on the concatenation of the shape and pathology concepts. Finally, a variational loss is computed using the parameters of these concept posteriors to achieve maximum a posteriori (MAP), and this loss is further combined with a position constraint loss and a cross-entropy loss to train the networks.

B. Hierarchical Semantic Concept Modeling

For K-class anatomy segmentation and N-class pathology segmentation, given a multisequence CMR image sampled from the multichannel variable $\boldsymbol{y} \in \mathbb{R}^{d_y \times C}$, where d_y denotes the dimension of $\boldsymbol{y}$, C denotes the number of sequences, and $\boldsymbol{y} = concat(I_{bSSFP}, I_{LGE}, I_{T2})$. Here, $concat(\cdot)$ denotes the channelwise concatenation, and I_{bSSFP}, I_{LGE}, and I_{T2} represent the bSSFP, LGE and T2 sequences, respectively. The image $\boldsymbol{y}$ is then decomposed into the sum of the concept shape $\boldsymbol{s} \in \mathbb{R}^{d_y \times 1}$, the concept pathology $\boldsymbol{o} \in \mathbb{R}^{d_y \times 1}$, and the concept appearance $\boldsymbol{a} \in \mathbb{R}^{d_y \times C}$, i.e., $\boldsymbol{y} = (\boldsymbol{s} + \boldsymbol{o} + \boldsymbol{a})$. Here, these three concept variables are related to the bSSFP, LGE, and T2 sequences. Then, the appearance $\boldsymbol{a}$ is modeled as domain-specific, following a Gaussian distribution determined by a mean $\boldsymbol{m} \in \mathbb{R}^{d_y \times C}$ and a covariance $diag(\boldsymbol{\rho})^{-1} \in \mathbb{R}^{(d_y \times d_y) \times C}$, i.e., $p(\boldsymbol{a}|\boldsymbol{m}, \boldsymbol{\rho}) = \mathcal{N}(\boldsymbol{a}|\boldsymbol{m}, diag(\boldsymbol{\rho})^{-1})$. Thus, the multisequence CMR images likelihood can be expressed as

$$p\left(\boldsymbol{y}|\boldsymbol{s}, \boldsymbol{o}, \boldsymbol{m}, \boldsymbol{\rho}\right) = \mathcal{N}\left(\boldsymbol{s} + \boldsymbol{o} + \boldsymbol{m}, diag\left(\boldsymbol{\rho}\right)^{-1}\right). \tag{1}$$

The semantic concept pathology $\boldsymbol{o}$ is modeled to be domain-invariant via a simultaneous autoregressive model (SAR) [56]. It is determined by an expected pathology segmentation $\boldsymbol{x} \in \mathbb{R}^{d_y \times N}$ and a boundary $\boldsymbol{\varepsilon} \in \mathbb{R}^{d_y}$ indicating edges of the pathology, namely,

$$p\left(\boldsymbol{o}|\boldsymbol{x}, \boldsymbol{\varepsilon}\right) = \prod_{n=1}^{N} \mathcal{N}\left(\boldsymbol{o}|\boldsymbol{0}, \left[\boldsymbol{D}_o^T \mathrm{diag}\left(\boldsymbol{x}_n \boldsymbol{\varepsilon}\right) \boldsymbol{D}_o\right]^{-1}\right) \tag{2}$$

where $\boldsymbol{x}_n$ denotes the pathology segmentation of the nth pathology class; $\boldsymbol{D}_o = \boldsymbol{I} - \boldsymbol{B}_o$ is a nonsingular matrix; $\boldsymbol{I}$ denotes an identity matrix; and each row of $\boldsymbol{B}_o \in \mathbb{R}^{d_y \times d_y}$ describes the interdependencies between a central pixel and its neighbors. For instance, the values of $\boldsymbol{B}_o$ for the four closest pixels are set to 0.25 while others are zeros, such that $\boldsymbol{D}_o$ is equal to the Laplacian operator.

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TABLE II
LIST OF CONJUGATE PRIORS AND VARIATIONAL POSTERIORS. NOTE THAT **1** REPRESENTS AN ALL-ONE VECTOR, AND $\boldsymbol{I}$ REPRESENTS AN IDENTITY MATRIX

Variable	Conjugate prior	Variational posterior
$\boldsymbol{m}$	$\mathcal{N}(\boldsymbol{m}\|\mu_m^{(0)}\mathbf{1}, (\sigma_m^{(0)})^{-1}\boldsymbol{I})$	$\mathcal{N}(\boldsymbol{m}\|\hat{\boldsymbol{\mu}}_m, \mathrm{diag}(\hat{\boldsymbol{\sigma}}_m^2))$
$\boldsymbol{\rho}$	$\prod_{i=1}^{d_y}\mathcal{G}(\rho_i\|\phi_{\rho i}^{(0)}, \gamma_{\rho i}^{(0)})$	$\prod_{i=1}^{d_y}\mathcal{G}(\rho_i\|\hat{\phi}_{\rho i}, \hat{\gamma}_{\rho i})$
$\boldsymbol{\upsilon}$	$\prod_{i=1}^{d_y}\mathcal{G}(\upsilon_i\|\phi_{\upsilon i}^{(0)}, \gamma_{\upsilon i}^{(0)})$	$\prod_{i=1}^{d_y}\mathcal{G}(\upsilon_i\|\hat{\phi}_{\upsilon i}, \hat{\gamma}_{\upsilon i})$
$\boldsymbol{\varepsilon}$	$\prod_{i=1}^{d_y}\mathcal{G}(\varepsilon_i\|\phi_{\varepsilon i}^{(0)}, \gamma_{\varepsilon i}^{(0)})$	$\prod_{i=1}^{d_y}\mathcal{G}(\varepsilon_i\|\hat{\phi}_{\varepsilon i}, \hat{\gamma}_{\varepsilon i})$
$\boldsymbol{\omega}$	$\prod_{i=1}^{d_y}\prod_{k=1}^{K}\mathcal{G}(\omega_{ki}\|\phi_{\omega ki}^{(0)}, \gamma_{\omega ki}^{(0)})$	$\prod_{i=1}^{d_y}\prod_{k=1}^{K}\mathcal{G}(\omega_{ki}\|\hat{\phi}_{\omega ki}, \hat{\gamma}_{\omega ki})$
$\boldsymbol{\eta}$	$\prod_{i=1}^{d_y}\prod_{n=1}^{N}\mathcal{G}(\eta_{ni}\|\phi_{\eta ni}^{(0)}, \gamma_{\eta ni}^{(0)})$	$\prod_{i=1}^{d_y}\prod_{n=1}^{N}\mathcal{G}(\eta_{ni}\|\hat{\phi}_{\eta ni}, \hat{\gamma}_{\eta ni})$
$\boldsymbol{\pi}$	$\prod_{k=1}^{K}\mathcal{B}(\pi_k\|\alpha_{\pi k}, \beta_{\pi k})$	$\prod_{k=1}^{K}\mathcal{B}(\pi_k\|\hat{\alpha}_{\pi k}, \hat{\beta}_{\pi k})$
$\boldsymbol{\tau}$	$\prod_{n=1}^{N}\mathcal{B}(\tau_n\|\alpha_{\tau n}, \beta_{\tau n})$	$\prod_{n=1}^{N}\mathcal{B}(\tau_n\|\hat{\alpha}_{\tau n}, \hat{\beta}_{\tau n})$

The segmentation $\boldsymbol{x}$ is modeled using another SAR, which is determined by the pathology segmentation boundary $\boldsymbol{\eta} \in \mathbb{R}^{d_y \times N}$ and a classwise probability $\boldsymbol{\tau} \in \mathbb{R}^N$, as follows:

$$p\left(\boldsymbol{x}|\boldsymbol{\tau},\boldsymbol{\eta}\right) = \prod_{n=1}^{N}\mathcal{N}\left(\boldsymbol{x}|\mathbf{0}, \left[-\ln\left(1-\boldsymbol{\tau}_n\right)\boldsymbol{D}_x^T\mathrm{diag}\left(\boldsymbol{\eta}_n\right)\boldsymbol{D}_x\right]^{-1}\right) \tag{3}$$

where the definition of $\boldsymbol{D}_x$ is similar to $\boldsymbol{D}_o$; $\boldsymbol{\eta}_n$ denotes the boundary of the nth segmentation $\boldsymbol{x}_n$; and $\boldsymbol{\tau}_n$ represents the probability that a pixel belongs to the nth class. In addition, the segmentation label $\boldsymbol{v} \in \mathbb{R}^{d_y \times N}$ is set to be a generalized Bernoulli distribution, i.e., $p(\boldsymbol{v}|\boldsymbol{x}) = \prod_{i=1}^{d_y}\prod_{n=1}^{N}\boldsymbol{x}_{in}^{v_{in}}$, where $\boldsymbol{x}$ denotes the pixelwise probability of the pathology segmentation prediction.

Similarly, the concept shape $\boldsymbol{s}$ and segmentation $\boldsymbol{z}$ are modeled by SARs in relation to their correlation variables, analogous to concept $\boldsymbol{o}$ and $\boldsymbol{x}$. For the specific form of the formula, please see the Supplementary Material.

Table II presents the selected prior distributions for $\boldsymbol{m}$, $\boldsymbol{\rho}$, $\boldsymbol{\varepsilon}$, $\boldsymbol{\upsilon}$, $\boldsymbol{\eta}$, $\boldsymbol{\omega}$, $\boldsymbol{\tau}$, and $\boldsymbol{\pi}$. Specifically, a Gaussian prior is assigned to $\boldsymbol{m}$, Gamma priors are assigned to $\boldsymbol{\rho}$, $\boldsymbol{\varepsilon}$, $\boldsymbol{\eta}$, $\boldsymbol{\upsilon}$, and $\boldsymbol{\omega}$, and Beta priors are assigned to $\boldsymbol{\tau}$ and $\boldsymbol{\pi}$. The choice of prior distributions for these concept variables will greatly simplify the subsequent derivations and computations. Through the previously mentioned hierarchical semantic concept modeling, HSCM-Net can provide domain-invariant concept variables and enhance the generalizability on MyoPS.

C. Inference of Variational Loss

To infer these concepts (i.e., $\boldsymbol{s}$, $\boldsymbol{a}$, $\boldsymbol{o}$, $\boldsymbol{z}$, and $\boldsymbol{x}$) from the multisequence CMR images cluster $\boldsymbol{y}$ and the corresponding segmentation label $\{\boldsymbol{u},\boldsymbol{v}\}$, a variational method is introduced. Let $\boldsymbol{\psi} = \{\boldsymbol{m},\boldsymbol{\rho},\boldsymbol{o},\boldsymbol{\varepsilon},\boldsymbol{x},\boldsymbol{\eta},\boldsymbol{\tau},\boldsymbol{s},\boldsymbol{\upsilon},\boldsymbol{z},\boldsymbol{\omega},\boldsymbol{\pi}\}$ be the set of all variables to be inferred. Our goal is to compute the posterior distribution $p(\boldsymbol{\psi}|\boldsymbol{y},\boldsymbol{u},\boldsymbol{v}) \propto p(\boldsymbol{y},\boldsymbol{u},\boldsymbol{v}|\boldsymbol{\psi})p(\boldsymbol{\psi})$ using MAP estimation. Given the infeasibility of direct computation, the VI approach is employed to approximate the posterior distribution $p(\boldsymbol{\psi}|\boldsymbol{y},\boldsymbol{u},\boldsymbol{v})$ by introducing a variational distribution $q(\boldsymbol{\psi})$. In the implementation of the VI approach, it is commonly assumed that the variables within the variational distribution $q(\boldsymbol{\psi})$ are independent of each other [57], namely (4), as shown at the bottom of the page.

Since a conjugate prior is assigned to each variable in $\boldsymbol{m}$, $\boldsymbol{\rho}$, $\boldsymbol{\varepsilon}$, $\boldsymbol{\upsilon}$, $\boldsymbol{\eta}$, $\boldsymbol{\omega}$, $\boldsymbol{\tau}$, and $\boldsymbol{\pi}$, the variational posterior retains the same form as its prior, as detailed in Table II. Similarly, for variables $\boldsymbol{o}$, $\boldsymbol{s}$, $\boldsymbol{x}$, and $\boldsymbol{z}$, their variational posteriors follow Gaussian distributions with the same form, e.g., $q(*) = \mathcal{N}(*|\hat{\mu}_*, \mathrm{diag}(\hat{\sigma}_*^2))$, $* \in \{\boldsymbol{o},\boldsymbol{s},\boldsymbol{x},\boldsymbol{z}\}$.

Then, the variational posterior parameters of these variables can be computed by minimizing the KL divergence between $q(\boldsymbol{\psi})$ and $p(\boldsymbol{\psi}|\boldsymbol{y},\boldsymbol{u},\boldsymbol{v})$, leading to

$$\begin{aligned}&\arg\min_{q(\boldsymbol{\psi})} KL\left(q\left(\boldsymbol{\psi}\right)\|p\left(\boldsymbol{\psi}|\boldsymbol{y},\boldsymbol{u},\boldsymbol{v}\right)\right)\\ &= \arg\min_{q(\boldsymbol{\psi})} KL\left(q\left(\boldsymbol{\psi}\right)\|p\left(\boldsymbol{\psi}\right)\right) - \mathbb{E}\left[\ln p\left(\boldsymbol{y},\boldsymbol{u},\boldsymbol{v}|\boldsymbol{\psi}\right)\right].\end{aligned} \tag{5}$$

Furthermore, the objective is transformed through reparameterization as follows:

$$\begin{aligned}*\underset{q(\boldsymbol{\psi})}{\mathrm{argmin}}\ &\underbrace{KL\left(q\left(\boldsymbol{\psi}\right)\|p\left(\boldsymbol{\psi}\right)\right) - \mathbb{E}_{q(\boldsymbol{\rho})}\left[\ln p\left(\boldsymbol{y}|\boldsymbol{s},\boldsymbol{o},\boldsymbol{m},\boldsymbol{\rho}\right)\right]}_{\mathcal{L}_{var}}\\ &\underbrace{-\mathbb{E}_{q(\boldsymbol{z})}\left[\ln p\left(\boldsymbol{u}|\boldsymbol{z}\right)\right]}_{\mathcal{L}_{CE_u}}\underbrace{-\mathbb{E}_{q(\boldsymbol{x})}\left[\ln p\left(\boldsymbol{v}|\boldsymbol{x}\right)\right]}_{\mathcal{L}_{CE_v}}.\end{aligned} \tag{6}$$

Note that minimizing the third and four terms of (6) induces two cross-entropy losses $\mathcal{L}_{CE_u}$ and $\mathcal{L}_{CE_v}$, corresponding to anatomy segmentation and pathology segmentation, respectively. Next, the focus is on minimizing the variational loss $\mathcal{L}_{var}$

$$*\underset{q(\boldsymbol{\psi})}{\mathrm{argmin}}\ \mathcal{L}_{var} = KL\left(q\left(\boldsymbol{\psi}\right)\|p\left(\boldsymbol{\psi}\right)\right) - \mathbb{E}_{q(\boldsymbol{\rho})}\left[\ln p\left(\boldsymbol{y}|\boldsymbol{\psi}\right)\right]. \tag{7}$$

Based on the PGM, the variational posteriors $q(\boldsymbol{\varepsilon})$, $q(\boldsymbol{\eta})$, $q(\boldsymbol{\tau})$, $q(\boldsymbol{\upsilon})$, $q(\boldsymbol{\omega})$, $q(\boldsymbol{\pi})$, and $q(\boldsymbol{\rho})$ in (7) can be sequentially minimized to derive explicit expressions for their corresponding parameters of the posterior distributions. The minimization of (7) over $q(\boldsymbol{\varepsilon})$ occurs when

$$\ln q\left(\boldsymbol{\varepsilon}\right) = \mathbb{E}_{o,x}\left[\ln p\left(\boldsymbol{o}|\boldsymbol{x},\boldsymbol{\varepsilon}\right)\right] + \ln p\left(\boldsymbol{\varepsilon}\right) + const \tag{8}$$

where *const* denotes a constant.

Solving the above equation yields

$$\hat{\boldsymbol{\mu}}_{\varepsilon} = \frac{2\boldsymbol{\gamma}_{\varepsilon}^{(0)} + N}{\sum_{n=1}^{N}\hat{\boldsymbol{\mu}}_{xn}\left[\left(\boldsymbol{D}_o\hat{\boldsymbol{\mu}}_o\right)^2 + 2\hat{\boldsymbol{\sigma}}_o\right] + 2\boldsymbol{\phi}_{\varepsilon}^{(0)}}. \tag{9}$$

The derived mean value of the concept pathology boundary $\boldsymbol{\varepsilon}$ will be lower for foreground pixels (higher $\hat{\boldsymbol{\mu}}_{xn}$) that have a stronger Laplacian response. Consequently, the pathology boundary highlights the transition area between the foreground and the background.

$$\begin{aligned}&q\left(\boldsymbol{\psi}\right)\\ &= q\left(\boldsymbol{m}\right)q\left(\boldsymbol{\rho}\right)q\left(\boldsymbol{o}\right)q\left(\boldsymbol{\varepsilon}\right)q\left(\boldsymbol{x}\right)q\left(\boldsymbol{\eta}\right)q\left(\boldsymbol{\tau}\right)q\left(\boldsymbol{s}\right)q\left(\boldsymbol{\upsilon}\right)q\left(\boldsymbol{z}\right)q\left(\boldsymbol{\omega}\right)q\left(\boldsymbol{\pi}\right).\end{aligned} \tag{4}$$

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TABLE III

LIST OF THE SPECIFIC FORMS AND DESCRIPTIONS OF EACH VARIATIONAL TERM IN THE VARIATIONAL LOSS EXPRESSED IN S (13). HERE, $\langle\cdot,\cdot\rangle$ DENOTES THE INNER PRODUCT, AND $\|\cdot\|^2_{\text{diag}(\cdot)}$ REPRESENTS THE SQUARED MAHALANOBIS DISTANCE

Modeling	Variational loss	Description
Image	$\mathcal{L}_y = \frac{1}{2}\|\boldsymbol{y} - (\boldsymbol{s}+\boldsymbol{m}+\boldsymbol{o})\|^2_{\text{diag}(\hat{\boldsymbol{\mu}}_\rho)}$	Facilitates the approximation of the observed CMR images $\boldsymbol{y}$
Pathology	$\mathcal{L}_{\hat{\mu}_o} = \frac{1}{2}\sum_{n=1}^{N}\|\boldsymbol{D}_o\hat{\boldsymbol{\mu}}_o\|^2_{\text{diag}(\hat{\boldsymbol{\mu}}_{xn}\hat{\boldsymbol{\mu}}_\varepsilon)}$ $\mathcal{L}_{\hat{\sigma}_o} = \frac{1}{2}\sum_{n=1}^{N}\langle 2\hat{\boldsymbol{\mu}}_{xn}\hat{\boldsymbol{\mu}}_\varepsilon, \hat{\boldsymbol{\sigma}}_o^2\rangle - \langle \mathbf{1}, \ln(\hat{\boldsymbol{\sigma}}_o^2)\rangle$	Ensures domain irrelevant $\hat{\boldsymbol{\mu}}_o$ with preserved edges Avoids the degradation of $q(\boldsymbol{o})$ into a one-point distribution
Pathology Segmentation	$\mathcal{L}_{\hat{\mu}_x} = \frac{1}{2}\sum_{n=1}^{N} c_n\|\boldsymbol{D}_x\hat{\boldsymbol{\mu}}_{xn}\|^2_{\text{diag}(\hat{\boldsymbol{\mu}}_{\eta n})}$ $\mathcal{L}_{\hat{\sigma}_x} = \frac{1}{2}\sum_{n=1}^{N} c_n\langle 2\hat{\boldsymbol{\mu}}_{\eta n}, \hat{\boldsymbol{\sigma}}_x^2\rangle - \langle \mathbf{1}, \ln(\hat{\boldsymbol{\sigma}}_x^2)\rangle$	Forces local concept $\hat{\boldsymbol{\mu}}_x$ with detected boundary Avoids the degradation of $q(\boldsymbol{x})$ into a one-point distribution
Appearance	$\mathcal{L}_{\hat{\mu}_m} = \frac{1}{2}\boldsymbol{\sigma}_m^{(0)}\|\hat{\boldsymbol{\mu}}_m\|_2^2$ $\mathcal{L}_{\hat{\boldsymbol{\sigma}}_m} = \frac{1}{2}[\langle \boldsymbol{\sigma}_m^{(0)}\mathbf{1}, \hat{\boldsymbol{\sigma}}_m^2\rangle - \langle \mathbf{1}, \ln(\hat{\boldsymbol{\sigma}}_m^2)\rangle]$	Constraints the energy of $\hat{\boldsymbol{\mu}}_m$ Avoids the degradation of $q(\boldsymbol{m})$ into a one-point distribution
Shape	$\mathcal{L}_{\hat{\mu}_s} = \frac{1}{2}\sum_{k=1}^{K}\|\boldsymbol{D}_s\hat{\boldsymbol{\mu}}_s\|^2_{\text{diag}(\hat{\boldsymbol{\mu}}_{zk}\hat{\boldsymbol{\mu}}_\upsilon)}$ $\mathcal{L}_{\hat{\sigma}_s} = \frac{1}{2}\sum_{k=1}^{K}\langle 2\hat{\boldsymbol{\mu}}_{zk}\hat{\boldsymbol{\mu}}_\upsilon, \hat{\boldsymbol{\sigma}}_s^2\rangle - \langle \mathbf{1}, \ln(\hat{\boldsymbol{\sigma}}_s^2)\rangle$	Ensures sequence irrelevant $\hat{\boldsymbol{\mu}}_s$ with preserved edges Avoids the degradation of $q(\boldsymbol{s})$ into a one-point distribution
Anatomy Segmentation	$\mathcal{L}_{\hat{\mu}_z} = \frac{1}{2}\sum_{k=1}^{K} c_k\|\boldsymbol{D}_z\hat{\boldsymbol{\mu}}_{zk}\|^2_{\text{diag}(\hat{\boldsymbol{\mu}}_{\omega k})}$ $\mathcal{L}_{\hat{\sigma}_z} = \frac{1}{2}\sum_{k=1}^{K} c_k\langle 2\hat{\boldsymbol{\mu}}_{\omega k}, \hat{\boldsymbol{\sigma}}_z^2\rangle - \langle \mathbf{1}, \ln(\hat{\boldsymbol{\sigma}}_z^2)\rangle$	Forces global concept $\hat{\boldsymbol{\mu}}_z$ with detected boundary Avoids the degradation of $q(\boldsymbol{z})$ into a one-point distribution

Similarly, minimizing (7) with respect to $q(\boldsymbol{\eta})$ results in the following formula:

$$\hat{\boldsymbol{\mu}}_{\eta n} = \frac{2\boldsymbol{\gamma}_{\eta n}^{(0)} + 1}{c_n\left[\left(\boldsymbol{D}_x\hat{\boldsymbol{\mu}}_{xn}\right)^2 + 2\hat{\boldsymbol{\eta}}_{xn}\right] + 2\boldsymbol{\phi}_{\eta n}} \tag{10}$$

where $c_n = \Psi(\hat{\alpha}_{\tau n} + \hat{\beta}_{\tau n}) - \Psi(\hat{\beta}_{\tau n})$, and $\Psi(\cdot)$ is the Digamma function.

Minimizing (7) with respect to $q(\boldsymbol{\tau})$ leads to the formula of computing $\hat{\alpha}_{\tau n}$, $\hat{\beta}_{\tau n}$

$$\begin{cases} \hat{\alpha}_{\tau n} = \alpha_{\tau n}^{(0)} + \dfrac{d_y}{2} \\ \hat{\beta}_{\tau n} = \dfrac{1}{2}\displaystyle\sum_{i=1}^{d_y}\boldsymbol{\mu}_{\eta n i}\left[\left(\boldsymbol{D}_s\hat{\boldsymbol{\mu}}_{sn}\right)_i^2 + 2\boldsymbol{\sigma}_{sni}^2\right] + \boldsymbol{\beta}_{\tau n}^{(0)}. \end{cases} \tag{11}$$

Finally, minimizing (7) with respect to $q(\boldsymbol{\rho})$ yields the computation formula for $\hat{\mu}_\rho$, which is given by

$$\hat{\boldsymbol{\mu}}_\rho = \frac{2\gamma_\rho^{(0)} + 1}{\left[\boldsymbol{y} - (\boldsymbol{s}+\boldsymbol{o}+\boldsymbol{m})\right]^2 + 2\phi_\rho^{(0)}}. \tag{12}$$

The induced mean value of variable $\boldsymbol{\rho}$ (i.e., the inverse variance of appearance) tends to be higher in regions with less noise and lower fitting error, thereby highlighting the noiseless areas. Then, the inference for $q(\boldsymbol{\varepsilon})$, $q(\boldsymbol{\eta})$, and $q(\boldsymbol{\tau})$ is similar to that for $q(\boldsymbol{\upsilon})$, $q(\boldsymbol{\omega})$, and $q(\boldsymbol{\pi})$.

After computing the parameters of $q(\boldsymbol{\varepsilon})$, $q(\boldsymbol{\eta})$, $q(\boldsymbol{\upsilon})$, $q(\boldsymbol{\omega})$, $q(\boldsymbol{\tau})$, $q(\boldsymbol{\pi})$, and $q(\boldsymbol{\rho})$, the remaining posterior distribution can be inferred. Therefore, the complete variational loss, which regularizes the appearance, pathology, shape, anatomy segmentation, and pathology segmentation of the multisequence CMR images, is summarized as

$$\mathcal{L}_{var} = \mathcal{L}_{\hat{\mu}_o} + \mathcal{L}_{\hat{\sigma}_o} + \mathcal{L}_{\hat{\mu}_x} + \mathcal{L}_{\hat{\sigma}_x} + \mathcal{L}_{var_y} \tag{13}$$

where $\mathcal{L}_{var_y}$ is the variational loss of other concept variables except $\boldsymbol{o}$ and $\boldsymbol{x}$, i.e., $\mathcal{L}_{var_y} = \mathcal{L}_y + \mathcal{L}_{\hat{\mu}_m} + \mathcal{L}_{\hat{\sigma}_m} + \mathcal{L}_{\hat{\mu}_s} + \mathcal{L}_{\hat{\sigma}_s} + \mathcal{L}_{\hat{\mu}_z} + \mathcal{L}_{\hat{\sigma}_z}$, and each variational term is summarized in Table III.

D. Total Loss

Scar and edema are located within the myocardial region, with scar typically positioned inside the areas of edema. This establishes a distinct spatial relationship among the MYO, edema, and scar. In order to fully leverage this prior spatial information to enhance the pathology segmentation capability of the model, MYO-pathology constraint loss and scar-edema constraint loss are introduced. Through these constraint losses, the model can better identify and segment scar and edema regions, thereby improving segmentation performance in unseen center domains.

Specifically, the MYO-pathology constraint loss consists of two components, MYO-scar loss ($\mathcal{L}_{con_{myoS}}$) and MYO-edema loss ($\mathcal{L}_{con_{myoE}}$), formulated as follows:

$$\mathcal{L}_{con_{myoS}} = -\sum_{i\in d_y}\left(1 - u_{myo}\right)\log\left(1 - x_{scar}\right) \tag{14}$$

$$\mathcal{L}_{con_{myoE}} = -\sum_{i\in d_y}\left(1 - u_{myo}\right)\log\left(1 - x_{edema}\right) \tag{15}$$

where u_{myo} is the label of binary myocardial segmentation, x_{scar} and x_{edema} are the predicted probability maps of scar and edema, respectively.

The scar-edema constraint loss also comprises two parts, edema-constrained scar loss ($\mathcal{L}_{con_S}$) and scar-constrained edema loss ($\mathcal{L}_{con_E}$), formulated as follows:

$$\mathcal{L}_{con_S} = -\frac{1}{Sum_{\text{edema}}}\sum_{i\in d_y}(1 - v_{edema})\log\left(1 - x_{scar}\right) \tag{16}$$

$$\mathcal{L}_{con_E} = -\frac{1}{Sum_{\text{scar}}}\sum_{i\in d_y} v_{scar}\log x_{edema} \tag{17}$$

where v_{scar} and v_{edema} are the segmentation labels of scar and edema, respectively. Sum_{edema} and Sum_{scar} are pathology volumes that ensure the robustness of the constraint loss against pathology regions of different sizes, expressed as $Sum_{\text{edema}} = \sum_v(1 - v_{edema})$ and $Sum_{\text{scar}} = \sum_v v_{scar}$.

Therefore, the total inclusiveness loss can be expressed as

$$\mathcal{L}_{con} = \lambda_1\mathcal{L}_{con_{myoS}} + \lambda_2\mathcal{L}_{con_{myoE}} + \lambda_3\left(\mathcal{L}_{con_S} + \mathcal{L}_{con_E}\right) \tag{18}$$

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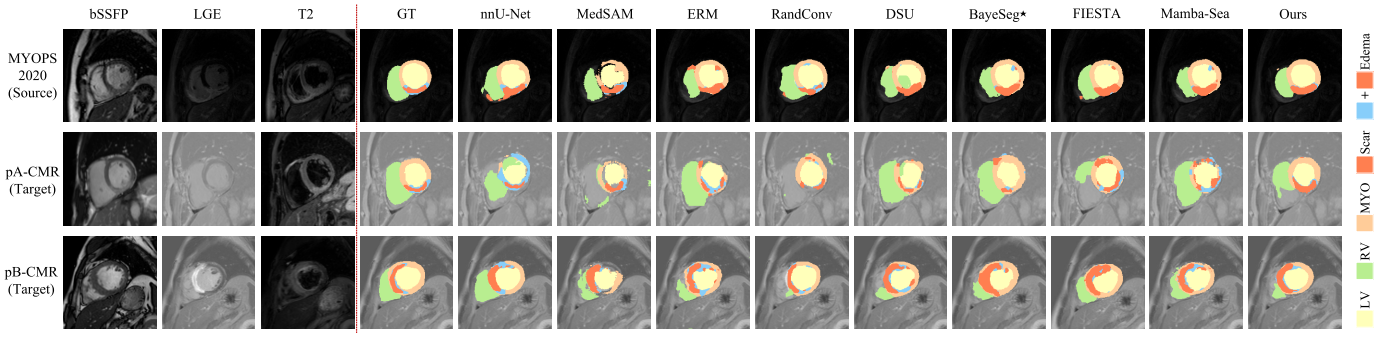


Fig. 3. Visualization of MyoPS results from different DG methods across three datasets, including the source domain and two unseen target domains. The results are overlaid on the LGE images. The labels for LV, RV, Myo, Scar, and Edema are, respectively, colored light yellow, light green, light orange, coral, and light blue.

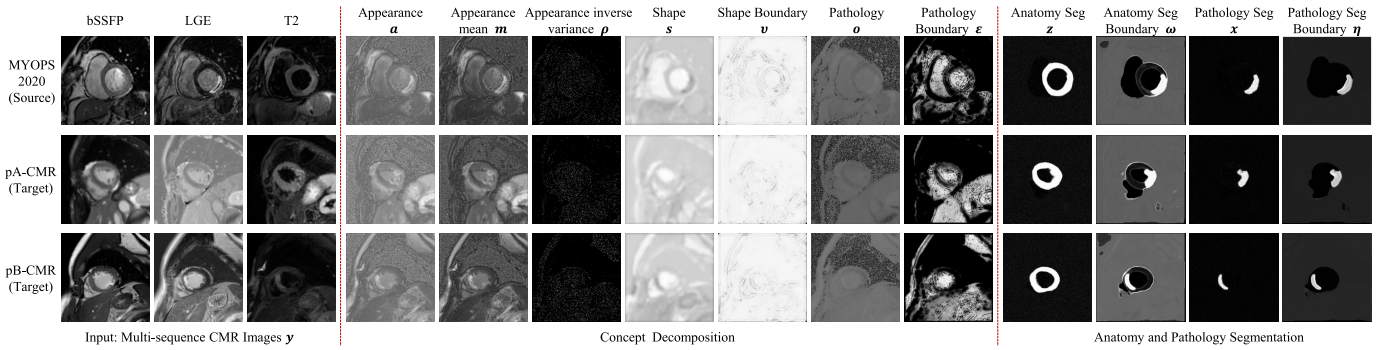


Fig. 4. Visualization of concept posteriors inferred by HSCM-Net. Here, "Seg": Segmentation.

where λ_1–λ_3 are three balancing parameters.

Combining the losses discussed above, the overall loss function for HSCM-Net is expressed as

$$\mathcal{L} = \mathcal{L}_{con} + \mathcal{L}_{CE_u} + \mathcal{L}_{CE_v} + \lambda_{var}\mathcal{L}_{var} \tag{19}$$

where λ_{var} is the balance weight for the variational loss.

E. Network Architecture

In the concept decomposition stage, three ResNet networks [58], denoted as $f_s(\cdot)$, $f_o(\cdot)$, and $f_a(\cdot)$ in Fig. 2(b), are employed to infer the variational posteriors for concept shape $\boldsymbol{s}$, pathology $\boldsymbol{o}$, and appearance mean $\boldsymbol{m}$, i.e., $q(\boldsymbol{s})$, $q(\boldsymbol{o})$, and $q(\boldsymbol{m})$, respectively. Note that the inputs to $f_a(\cdot)$ and $f_o(\cdot)$ are given by residual $(\boldsymbol{y}-\boldsymbol{s})$. The networks $f_s(\cdot)$ and $f_o(\cdot)$ each contain ten residual blocks, with the structure of each block being "Conv + ReLU + Conv." Both $f_s(\cdot)$ and $f_o(\cdot)$ produce outputs with two channels, the first channel represents the pixelwise mean of the shape $(\hat{\boldsymbol{\mu}}_x)$ and the pathology $(\hat{\boldsymbol{\mu}}_o)$, and the other channel represents their pixelwise variances $\hat{\boldsymbol{\sigma}}_x$ and $\hat{\boldsymbol{\sigma}}_o$. The shape concept $\boldsymbol{s}$ and pathology concept $\boldsymbol{o}$ represent random samples from $q(\boldsymbol{s})$ and $q(\boldsymbol{o})$, respectively. The network $f_a(\cdot)$ contains six residual blocks, with the structure of each block being "Conv + BN + ReLU + Conv + BN." In a similar manner, $f_a(\cdot)$ outputs the mean and variance for the $q(\boldsymbol{m})$. By sampling $\boldsymbol{m}$ and computing $\boldsymbol{\rho}$, the appearance $\boldsymbol{a}$ can be stochastically sampled from the Gaussian distribution $\mathcal{N}(\boldsymbol{m}, diag(\hat{\boldsymbol{\mu}}_\rho)^{-1})$.

In the segmentation stage, a U-Net [59] is employed as the network backbone of $g(\cdot)$ to simultaneously infer the variational posterior of both segmentations $\boldsymbol{z}$ and $\boldsymbol{x}$, where $\boldsymbol{s}$ and $\boldsymbol{o}$ are concatenated as the input of this network to leverage their spatial positional relationship. The output of $g(\cdot)$ has $2 * (K + N)$ channels, with the first $K + N$ channels representing the pixelwise means of the anatomy and pathology segmentations, and the remaining channels representing their pixelwise variances. It is important to note that $\boldsymbol{z}$ and $\boldsymbol{x}$ are random samples from the obtained posterior distribution, treated as random segmentations used for training. During the testing phase, the predicted segmentations are obtained by taking the means of $\boldsymbol{z}$ and $\boldsymbol{x}$, as the variational posteriors $q(\boldsymbol{z})$ and $q(\boldsymbol{x})$ are Gaussian distributions with modes equal to their means.

IV. Experiments

A. Datasets and Experiment Settings

The proposed HSCM-Net was evaluated on three-domain datasets from different centers, including two private multisequence CMR datasets (pA-CMR and pB-CMR datasets) and one public multisequence CMR dataset (the MYOPS2020 challenge dataset [60]).

1) pA-CMR Dataset: This private dataset comprises 50 CMR images from three imaging sequences (bSSFP, LGE, and T2), collected from a hospital in China. These images were acquired using a Siemens Aera 1.5T scanner with the following imaging parameters. For the bSSFP sequence, the field of view (FOV) was 276–453 × 283–453 mm^2, in-plane resolution was 1.48–2.36 × 1.48–2.36 mm^2, the number of slices ranged from 8 to 10, and slice thickness was 10 mm. For the LGE sequence, the FOV was 276–453 ×283–453 mm^2, in-

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plane resolution was 1.33–1.86 × 1.33–1.86 mm^2, the number of slices ranged from 4 to 11, and slice thickness was 10 mm. For the T2 sequence, the FOV was 276–453 × 283–453 mm^2, in-plane resolution was 1.11–1.77 × 1.11–1.77 mm^2, the number of slices ranged from 2 to 10, and slice thickness was 10 mm. All images were aligned to a common space using the MvMM method [61], [62]. Annotations for five cardiac structures were provided by a cardiologist, including the LV cavity, right ventricle (RV) cavity, LV MYO, scar, and edema.

2) pB-CMR Dataset: This private dataset includes 50 multisequence CMR images from another hospital in China, comprising the same three imaging sequences, i.e., bSSFP, LGE, and T2. All images were obtained using a Philips Ingenia 1.5T scanner with the following acquisition parameters. For the bSSFP sequence, the FOV was 300 × 300 mm^2, in-plane resolution was 1.17 × 1.17 mm^2, the number of slices ranged from 10 to 20, and slice thickness was 6 mm. For both the LGE and T2 sequences, the FOV was 300 × 300 mm^2, in-plane resolution was 0.89 × 0.89 mm^2, slice thickness was 10 mm, and the number of slices ranged from 3 to 12 (LGE) and 3 to 11 (T2), respectively. Similar to the pA-CMR, all images were aligned to a common space using the MvMM method. The same five cardiac structures were annotated by a cardiologist to provide gold-standard labels.

3) MYOPS2020 Challenge Dataset: This dataset contains 45 aligned multisequence CMR (i.e., bSSFP, LGE, and T2) images, comprising 25 labeled samples and 20 unlabeled samples. These CMR images are also manually annotated with five structures.

The MYOPS2020 dataset was used as the source domain, with 20 labeled cases for training, five labeled cases for validation, and 20 unlabeled cases for testing (evaluated using official tools). All cases from the pA-CMR and pB-CMR datasets were used as unseen target domains for testing. All samples were resampled into 2-D slices with a resolution of 1 × 1 mm^2 and cropped to 196 × 196 pixels with the heart as the center. During the training process, normalization was performed using Z-scores. To reduce overfitting, data augmentation strategies were applied, including random rotation (0°−360°), random flipping (probability of 0.5), and random scaling (factors 0.8−1.2). The Adam optimizer was employed to update network parameters, with the learning rate set to 0.0001. Furthermore, the prior hyperparameter settings for the PGM in Fig. 2(a) were as follows: $\boldsymbol{\alpha}_\tau^{(0)} = \boldsymbol{\alpha}_\pi^{(0)} = 2$ and $\boldsymbol{\beta}_\tau^{(0)} = \boldsymbol{\beta}_\pi^{(0)} = 2$; $\boldsymbol{\mu}_m^{(0)} = 0$ and $\boldsymbol{\sigma}_m^{(0)} = 1$. The values of $\boldsymbol{\gamma}_*^{(0)}$ were set to 2, and $\boldsymbol{\phi}_\rho^{(0)}$, $\boldsymbol{\phi}_\varepsilon^{(0)}$, $\boldsymbol{\phi}_\eta^{(0)}$ $\boldsymbol{\phi}_\upsilon^{(0)}$, and $\boldsymbol{\phi}_\omega^{(0)}$ were set to 10^{-6}, 10^{-6}, 10^{-3}, 10^{-6}, and 10^{-2}, respectively. The required parameters, such as $\hat{\boldsymbol{\mu}}_\varepsilon$ and $\hat{\boldsymbol{\mu}}_\upsilon$ used in $\mathcal{L}_{var}$, were computed as illustrated in Fig. 2(b). In our experiments, the balance weights λ_1, λ_2, λ_3, and λ_{var} were set to 1, 0.1, 0.2, and 100, respectively. All tasks were conducted on a single 24-GB NVIDIA RTX 3090 GPU, and after 2000 epochs of training, the best model was selected from the validation set for evaluation on the testing set.

B. Comparison Study

To evaluate the generalizability of the proposed method, HSCM-Net was compared with two categories of approaches: 1) two representative standard medical image segmentation methods, nnU-Net [63] and MedSAM [8]; and 2) six other DG methods, including ERM [64], RandConv [65], DSU [10], BayeSeg*[1] [9], FIESTA [7], and Mamba-Sea [6]. Notably, the first four DG segmentation methods and ours were trained under a unified framework with the same backbone.

Table IV presents the results, including the Dice score drop (↓) compared with source domain performance. Overall, in the target domains, HSCM-Net achieves an average Dice score of 0.577 and 0.695 for pathology segmentation and five structures segmentation, respectively, outperforming the second-best method (i.e., DSU) by 0.074 ($p < 0.05$ and $T_W = 1092$)[2] and 0.034 ($p < 0.05$ and $T_W = 1481$), respectively. In the target domain, the pACMR dataset, our method outperforms the second-best method by a significant margin of 0.104 ($p < 0.05$ and $T_t = 7.38$)[3] in the challenging pathology segmentation task, with a small generalizability drop of 0.036. In another target domain, the pB-CMR dataset, HSCM-Net achieves slightly higher Dice scores than the second-best method by 0.046 ($p = 0.35$ and $T_W = 541$) and 0.017 ($p = 0.39$ and $T_W = 549$) in pathology segmentation and five-structures segmentation, respectively, with the smallest performance drop of 0.015 and 0.014. These results show that HSCM-Net achieves the best performance and generalizability, which confirms the effectiveness of hierarchical semantic concept modeling. For qualitative comparisons, representative cases from the three-domain CMR datasets are visualized in Fig. 3. Compared with other methods, our HSCM-Net reduces segmentation errors in scar and edema, achieves smoother and more accurate boundary delineations for the five structures, and produces segmentation results that are closer to the gold standard.

C. Ablation Study

To validate the effectiveness of the stochastic mapping (SM, i.e., the randomness caused by variable resampling), the variational loss ($\mathcal{L}_{var}$) and the position constraint loss ($\mathcal{L}_{con}$) in HSCM-Net, three distinct ablation experiments were conducted. The configurations for these experiments are delineated in Table V, where the symbols "×" and "✓," respectively, indicate the exclusion and inclusion of the specified setting.

From Table V, compared with Model #4, the generalizability of Model #1 (without SM and $\mathcal{L}_{var}$) is degraded, with the MyoPS Dice score decreasing from 0.567 to 0.504 on the target pA-CMR dataset. Model #2 utilized the SM strategy but did not incorporate $\mathcal{L}_{var}$, resulting in lower pathology segmentation accuracy on the target pA-CMR and pB-CMR datasets compared with Model #1. These two comparisons demonstrate that SM and variational loss are pivotal to the model generalizability, with the introduction of SM alone insufficient for enhancing performance. SM serves as a prerequisite for the application of variational loss. When guided

[1]BayeSeg* is a two-stage DG method. First, it uses the raw BayeSeg for anatomy segmentation, and then applies U-Net to segment pathology regions.

[2]T_W: Test statistic value of Wilcoxon signed-rank test.

[3]T_t: Test statistic value of paired Student's t-test.

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TABLE IV

PERFORMANCE COMPARISON WITH DIFFERENT DG METHODS. NOTE THAT THE SYMBOL "↓" DENOTES THE DECREASE IN DICE SCORE COMPARED WITH THE SOURCE DOMAIN. "MYOPS" IS THE AVERAGE OF THE DICE VALUES FOR SCAR SEGMENTATION AND EDEMA SEGMENTATION. "AVG." IS THE AVERAGE OF THE DICE VALUES FOR THE SEGMENTATION OF THE FIVE STRUCTURES (I.E., LV, RV, MYO, SCAR, AND EDEMA). "TS": TARGETS. THE BEST PERFORMANCES ARE MARKED IN BOLD

Method	MYOPS2020 (Source)							pA-CMR (Target)							pB-CMR (Target)								
	LV	RV	MYO	Scar	Edema	MyoPS	Avg.	LV	RV	MYO	Scar	Edema	MyoPS	Avg.	LV	RV	MYO	Scar	Edema	MyoPS	Avg.	MyoPS (Ts)	Avg. (Ts)
nnU-Net [63]	.918 ±.022	.887 ±.051	.872 ±.042	.597 ±.232	.685 ±.109	.641	.792	.817 ±.113	.668 ±.197	.721 ±.132	.363 ±.182	.570 ±.174	.466 ↓.175	.628 ↓.164	.859 ±.142	.712 ±.255	.776 ±.164	.501 ±.184	.563 ±.193	.532 ↓.109	.682 ↓.110	.499	.655
MedSAM [8]	.812 ±.062	.785 ±.097	.687 ±.113	.548 ±.231	.639 ±.118	.584	.694	.861 ±.058	.472 ±.215	.440 ±.182	.429 ±.152	.466 ±.171	.447 ↓.137	.534 ↓.160	.876 ±.034	.615 ±.207	.569 ±.184	.485 ±.155	.547 ±.132	.516 ↓.068	.618 ↓.076	.482	.576
ERM [64]	.878 ±.072	.804 ±.130	.835 ±.082	.568 ±.220	.586 ±.142	.577	.734	.775 ±.162	.624 ±.129	.694 ±.141	.354 ±.183	.494 ±.171	.424 ↓.153	.588 ↓.146	.877 ±.060	.732 ±.197	.799 ±.089	.473 ±.152	.538 ±.155	.505 ↓.072	.684 ↓.050	.465	.636
RandConv [65]	.894 ±.082	.765 ±.214	.833 ±.088	.537 ±.235	.552 ±.143	.545	.716	**.905 ±.031**	.086 ±.229	.757 ±.105	.368 ±.145	.548 ±.152	.458 ↓.087	.533 ↓.183	.882 ±.048	.359 ±.384	.777 ±.118	.456 ±.187	.513 ±.187	.484 ↓.061	.597 ↓.119	.471	.565
DSU [10]	.878 ±.082	.775 ±.192	.842 ±.085	.561 ±.223	.591 ±.148	.576	.729	.774 ±.179	.645 ±.154	.746 ±.102	.415 ±.140	.512 ±.155	.463 ↓.113	.618 ↓.111	.881 ±.070	.748 ±.188	.801 ±.080	.518 ±.141	.566 ±.160	.542 ↓.034	.703 ↓.026	.503	.661
BayeSeg* [9]	.902 ±.028	.841 ±.108	.872 ±.031	.481 ±.215	.551 ±.132	.516	.730	.899 ±.045	**.771 ±.118**	**.812 ±.053**	.284 ±.180	.456 ±.195	.370 ↓.146	.645 ↓.085	.911 ±.023	**.793 ±.182**	.835 ±.043	.383 ±.122	.471 ±.127	.427 ↓.089	.678 ↓.052	.399	.662
FIESTA [7]	.893 ±.073	.801 ±.223	.864 ±.051	.616 ±.217	.647 ±.155	.632	.764	.855 ±.095	.314 ±.253	.771 ±.105	.393 ±.137	.474 ±.146	.433 ↓.199	.561 ↓.203	.888 ±.033	.625 ±.286	.828 ±.042	.515 ±.113	.551 ±.116	.533 ↓.099	.682 ↓.082	.483	.622
Mamba-Sea [6]	**.918 ±.031**	**.894 ±.059**	**.881 ±.034**	**.647 ±.235**	**.707 ±.101**	**.677**	**.809**	.764 ±.137	.681 ±.151	.651 ±.149	.307 ±.141	.477 ±.152	.392 ↓.285	.576 ↓.233	.875 ±.072	.741 ±.191	.811 ±.070	.539 ±.126	.591 ±.124	.565 ↓.112	.711 ↓.098	.479	.644
Ours	.888 ±.047	.724 ±.220	.848 ±.053	.576 ±.236	.631 ±.134	.603	.733	.877 ±.078	.543 ±.212	.803 ±.080	**.516 ±.109**	**.618 ±.135**	**.567 ↓.036**	**.672 ↓.061**	**.915 ±.028**	.659 ±.227	**.846 ±.033**	**.551 ±.101**	**.624 ±.104**	**.588 ↓.015**	**.719 ↓.014**	**.577**	**.695**

TABLE V

EFFECTS OF SM, VARIATIONAL LOSS ($\mathcal{L}_{var}$), AND CONSTRAINT LOSS ($\mathcal{L}_{con}$) ON HSCM-NET. NOTE THAT THE SYMBOL "↓" DENOTES THE DECREASE IN DICE SCORE COMPARED WITH THE SOURCE DOMAIN. "MYOPS" IS THE AVERAGE OF THE DICE VALUES FOR SCAR AND EDEMA SEGMENTATIONS. "AVG." IS THE AVERAGE OF THE DICE VALUES FOR THE SEGMENTATION OF THE FIVE STRUCTURES (I.E., LV, RV, MYO, SCAR, AND EDEMA). THE BEST PERFORMANCES ARE MARKED IN BOLD

Model				MYOPS2020 (Source)							pA-CMR (Target)							pB-CMR (Target)						
	SM	$\mathcal{L}_{var}$	$\mathcal{L}_{con}$	LV	RV	MYO	Scar	Edema	MyoPS	Avg.	LV	RV	MYO	Scar	Edema	MyoPS	Avg.	LV	RV	MYO	Scar	Edema	MyoPS	Avg.
#1	×	×	✓	.904 ±.027	**.825 ±.126**	.848 ±.049	.554 ±.218	.630 ±.108	.592	**.752**	.874 ±.058	.639 ±.147	.752 ±.068	.426 ±.118	.582 ±.146	.504 ↓.088	.654 ↓.098	.906 ±.027	.709 ±.193	.820 ±.043	.529 ±.096	**.628 ±.085**	.578 ↓.014	.718 ↓.034
#2	✓	×	✓	**.910 ±.025**	.802 ±.113	**.858 ±.041**	**.580 ±.226**	.595 ±.128	.587	.749	.868 ±.086	**.645 ±.169**	.781 ±.082	.402 ±.118	.566 ±.133	.484 ↓.103	.653 ↓.096	.906 ±.033	**.743 ±.176**	.825 ±.049	.531 ±.104	.575 ±.094	.553 ↓.034	.716 ↓.033
#3	✓	✓	×	.905 ±.043	.756 ±.218	.846 ±.057	.460 ±.196	.569 ±.120	.514	.707	**.894 ±.064**	.492 ±.201	.784 ±.099	.430 ±.095	.606 ±.143	.518 ↑.004	.641 ↓.066	.908 ±.041	.615 ±.255	.826 ±.061	.489 ±.119	.586 ±.124	.537 ↑.023	.685 ↓.022
#4 (ours)	✓	✓	✓	.888 ±.047	.724 ±.220	.848 ±.053	.576 ±.236	**.631 ±.134**	**.603**	.733	.877 ±.078	.543 ±.212	**.803 ±.080**	**.516 ±.109**	**.618 ±.135**	**.567 ↓.036**	**.672 ↓.061**	**.915 ±.028**	.659 ±.227	**.846 ±.033**	**.551 ±.101**	.624 ±.104	**.588 ↓.015**	**.719 ↓.014**

by the variational loss, SM enables the model to perceive an infinite number of samples related to the shape and pathology aspects of given multisequence CMR images, thus aiding in the enhancement of generalizability on unseen domains. Moreover, without the guidance of the variational loss, the concept modeling-driven fusion strategy fails, leading to a degradation in the model generalizability. In addition, the incorporation of $\mathcal{L}_{con}$ substantially improves the segmentation accuracy of pathology, increasing the MyoPS Dice score by 0.049 and 0.051 on the pA-CMR and pB-CMR datasets, respectively. Notably, for the challenging task of scar segmentation, the Dice score improves considerably from 0.430 to 0.516 on the pA-CMR dataset. This improvement is attributed to the fact that scar and edema regions are interrelated and both are located in the MYO.

D. Visualization of Concept Variables

To verify the effectiveness of the concept modeling method, the posterior distributions of the concept variables were directly visualized. As shown in Fig. 4, during the concept decomposition stage, the input multisequence CMR images are successfully decomposed into shape, appearance, and pathology concepts. The concepts $\boldsymbol{s}$ and $\boldsymbol{o}$ are modeled as domain-invariant variables, capturing global anatomy and local pathology information, respectively. Consequently, $\boldsymbol{\upsilon}$ and $\boldsymbol{\varepsilon}$ are introduced to detect the boundaries of anatomy and pathology, respectively. As illustrated in the eighth column of Fig. 4, $\boldsymbol{\upsilon}$ successfully delineates the boundary of the LV MYO, which is difficult to identify in the original multisequence CMR images. In addition, in the tenth column of Fig. 4, $\boldsymbol{\varepsilon}$ prominently displays the myocardial pathology region, which is challenging to discern in the input CMR images. During the segmentation stage, the $\boldsymbol{s}$ and $\boldsymbol{o}$ concept variables are concatenated to segment the five structures. To improve segmentation accuracy around the boundaries, $\boldsymbol{\omega}$ and $\boldsymbol{\eta}$ are assigned to detect the boundaries of anatomy and pathology, respectively. These variables can effectively indicate the segmentation boundaries of the target structures. The visualization of these posterior distributions confirms that the proposed method accurately models both domain-invariant and domain-specific concepts, thereby improving the generalizability of the model.

To quantitatively validate the independence of the decomposed semantic concepts, the normalized mutual information (NMI) was calculated among the three latent concept variables: appearance $\boldsymbol{a}$, pathology $\boldsymbol{o}$, and shape $\boldsymbol{s}$. As presented in Table VI, consistently low NMI scores were observed across all three datasets (i.e., MyoPS2020, pA-CMR, and pB-CMR),

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TABLE VI

QUANTITATIVE ANALYSIS OF CONCEPT INDEPENDENCE VIA NMI. THE NMI SCORES ARE CALCULATED BETWEEN PAIRS OF LATENT CONCEPT VARIABLES: APPEARANCE ($\boldsymbol{a}$), PATHOLOGY ($\boldsymbol{o}$), AND SHAPE ($\boldsymbol{s}$) ACROSS THE SOURCE (MYOPS2020) AND TARGET (PA-CMR AND PB-CMR) DATASETS. "AVG." DENOTES THE AVERAGE NMI VALUE

Dataset	NMI($\boldsymbol{a}$, $\boldsymbol{s}$)	NMI($\boldsymbol{a}$, $\boldsymbol{o}$)	NMI($\boldsymbol{s}$, $\boldsymbol{o}$)	Avg. NMI
MyoPS2020	0.0457	0.1399	0.0838	0.0898
pA-CMR	0.0544	0.1071	0.0908	0.0841
pB-CMR	0.0585	0.1284	0.0998	0.0956
Mean	0.0529	0.1251	0.0915	0.0898

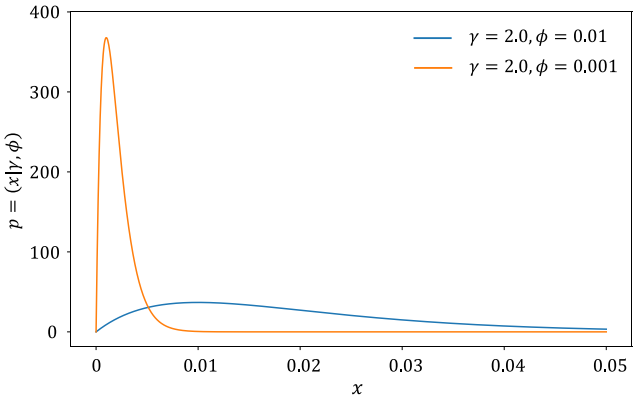


Fig. 5. PDF of the Gamma distribution with different parameters.

resulting in an overall mean NMI of 0.0898. Specifically, the low interaction between $\boldsymbol{a}$ and $\boldsymbol{s}$ (mean NMI = 0.0529) as well as $\boldsymbol{s}$ and $\boldsymbol{o}$ (mean NMI = 0.0915) quantitatively demonstrates that these variables share minimal information. In addition, the NMI between $\boldsymbol{s}$ and $\boldsymbol{o}$ (mean NMI = 0.1251), while exhibiting a slight increase, remains at a limited level. These quantitative results confirm that HSCM-Net effectively decomposes multisequence CMR images into three relatively independent concepts, providing support that complements the qualitative visualizations.

E. Generalizability of the Concept Modeling

1) Study of the Gamma Prior $\boldsymbol{\omega}$ and $\boldsymbol{\eta}$: The impact of the Gamma prior distributions $\boldsymbol{\omega}$ and $\boldsymbol{\eta}$ on hierarchical semantic concept modeling was investigated. The segmentation boundaries $\boldsymbol{\omega}$ and $\boldsymbol{\eta}$ reflect the spatial correlation between pixels within the anatomical structure region $\boldsymbol{z}$ and the pathology region $\boldsymbol{x}$, while also capturing the segmentation uncertainty and scale characteristics of $\boldsymbol{z}$ and $\boldsymbol{x}$. The results in Table VII show that varying the hyperparameters of the priors affects segmentation performance across different target regions. Specifically, smaller $\boldsymbol{\phi}$ (scale) values of Gamma priors are beneficial for modeling local information, improving generalizability in small-area pathology segmentation. Conversely, larger $\boldsymbol{\phi}$ values of Gamma priors are advantageous for modeling global information, leading to better segmentation of large-area anatomical structures. The probability density function (pdf) curves in Fig. 5 further support this observation. Larger values of $\boldsymbol{\phi}$ result in higher means and variances, yielding more dispersed probability distributions favorable for capturing global information. Conversely, smaller values of $\boldsymbol{\phi}$ have lower means and variances, leading to more concentrated probability distributions, representing a focus on local information. Thus, selecting appropriate prior parameters for $\boldsymbol{\omega}$ and $\boldsymbol{\eta}$ is crucial for achieving optimal model generalizability. The proposed prior parameter settings, i.e., $\boldsymbol{\phi}_{\omega}^{(0)} = 10^{-2}$, $\boldsymbol{\gamma}_{\omega}^{(0)} = 2$, $\boldsymbol{\phi}_{\eta}^{(0)} = 10^{-3}$, and $\boldsymbol{\gamma}_{\eta}^{(0)} = 2$, yield the best overall performance.

2) Study of Different PGM Configurations: To explore the benefits of hierarchical concept in statistical modeling, two simplified PGMs were designed to assess HSCM-Net. In one configuration, the anatomy segmentation variable $\boldsymbol{z}$ was set to be deterministic rather than sampled from the posterior distribution. This modification impacts both the variational loss and the SM in the concept modeling. Specifically, the corresponding output channels of network $g(\cdot)$ are halved, directly extracting the precise value of the anatomy segmentation. In addition, $\boldsymbol{\omega}$ and $\boldsymbol{\pi}$ were pruned together, simplifying the variational loss and eliminating the need for $\mathcal{L}_{\hat{\mu}_z}$ and $\mathcal{L}_{\hat{\sigma}_z}$. In a similar fashion, another PGM configuration set the pathology segmentation variable $\boldsymbol{x}$ to be deterministic, which led to corresponding modifications in the $g(\cdot)$ and the variational loss.

Table VIII shows the comprehensive performance of HSCM-Net across various PGM configurations. As seen, Model #7 of HSCM-Net with complete PGM achieves the best performance among all configurations. The results from Model #1, #2 and #3 reveal that enforcing determinism on $\boldsymbol{z}$ diminishes the model generalizability, especially for anatomy segmentation, because the variable $\boldsymbol{z}$ is essential for capturing sequence-irrelevant anatomy concept information. Similarly, the results of Models #4–#6 indicate that setting variable $\boldsymbol{x}$ to be deterministic also leads to a decline in model generalizability, due to the absence of semantic concept modeling.

3) Study of Modeling Pathology Concept: To further analyze the effectiveness of modeling pathology semantic concept on model generalizability, two BayeSeg variant models (i.e., BayeSeg+ and BayeSeg++) were designed and compared. BayeSeg+ directly segmented two pathology regions on the shape variable, while BayeSeg++ segmented the pathology regions on the appearance variable. The experimental results from Table IX reveal deficiencies in both BayeSeg+ and BayeSeg++ regarding the accuracy and generalizability of pathology segmentation. This suggests that modeling domain-invariant pathology information on either the shape or appearance variable does not yield satisfactory model generalizability. This is because the BayeSeg framework relies on explicit prior rules suited for stable organ structures, making it unsuitable for handling pathology regions that exhibit variability in size, shape, and morphology or may be absent. Conversely, our proposed hierarchical semantic concept modeling decomposes multisequence CMR images into three concept variables, with a particular emphasis on the global anatomy shape concept and local high-level semantic pathology concept, resulting in improved model generalizability. This is because the concept modeling method effectively models the concept distribution of domain-invariant pathology information, segmenting the target regions based on this distribution.

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TABLE VII

RESULTS OF THE AVERAGE DICE SCORES FOR DIFFERENT GAMMA PRIORS OF $\boldsymbol{\omega}$ AND $\boldsymbol{\eta}$. NOTE THAT THE PRIOR SETTINGS FOR OTHER VARIABLES ARE KEPT CONSTANT

Model	Prior of $\boldsymbol{\omega}$		Prior of $\boldsymbol{\eta}$		MYOPS2020 (Source)							pA-CMR (Target)							pB-CMR (Target)						
	$\phi_\omega^{(0)}$	$\gamma_\omega^{(0)}$	$\phi_\eta^{(0)}$	$\gamma_\eta^{(0)}$	LV	RV	MYO	Scar	Edema	MyoPS	Avg.	LV	RV	MYO	Scar	Edema	MyoPS	Avg.	LV	RV	MYO	Scar	Edema	MyoPS	Avg.
#1	10^{-2}	2	10^{-2}	2	.876 ±.070	**.794 ±.167**	.815 ±.080	.536 ±.222	.607 ±.128	.571	.726	.874 ±.074	**.569 ±.145**	.729 ±.096	.431 ±.126	.556 ±.148	.493 ↓.075	.632 ↓.094	.899 ±.030	.647 ±.207	.797 ±.048	.448 ±.134	.580 ±.117	.514 ↓.057	.674 ↓.052
#2	10^{-3}	2	10^{-3}	2	**.908 ±.018**	.682 ±.241	**.860 ±.042**	.526 ±.229	.612 ±.137	.569	.718	**.903 ±.032**	.322 ±.209	.773 ±.073	.441 ±.129	.579 ±.177	.510 ↓.059	.604 ↓.114	.905 ±.032	.594 ±.233	.819 ±.057	.504 ±.101	**.630 ±.123**	.567 ↓.002	.690 ↓.028
#3	10^{-3}	2	10^{-2}	2	.869 ±.061	.749 ±.193	.844 ±.050	.552 ±.233	.625 ±.123	.589	.728	.882 ±.076	.558 ±.185	.773 ±.054	.456 ±.101	.616 ±.118	.536 ↓.053	.657 ↓.071	.892 ±.031	.657 ±.247	.791 ±.051	.484 ±.109	.584 ±.110	.534 ↓.055	.682 ↓.046
#4 (ours)	10^{-2}	2	10^{-3}	2	.888 ±.047	.724 ±.220	.848 ±.053	**.576 ±.236**	**.631 ±.134**	**.603**	**.733**	.877 ±.078	.543 ±.212	**.803 ±.080**	**.516 ±.109**	**.618 ±.135**	**.567 ↓.036**	**.672 ↓.061**	**.915 ±.028**	**.659 ±.227**	**.846 ±.033**	**.551 ±.101**	.624 ±.104	**.588 ↓.015**	**.719 ↓.014**

TABLE VIII

RESULTS OF THE AVERAGE DICE SCORES FOR DIFFERENT PGM CONFIGURATIONS. NOTE THAT "RM" REFERS TO THE RANDOM MAPPING OF THE RELATED CNN, WHILE "L_v" REPRESENTS THE VARIATIONAL LOSS OF THE RELATED VARIABLE

Model	Stochastic $\boldsymbol{z}$		Stochastic $\boldsymbol{x}$		MYOPS2020 (Source)							pA-CMR (Target)							pB-CMR (Target)						
	RM	L_v	RM	L_v	LV	RV	MYO	Scar	Edema	MyoPS	Avg.	LV	RV	MYO	Scar	Edema	MyoPS	Avg.	LV	RV	MYO	Scar	Edema	MyoPS	Avg.
#1	×	✓	✓	✓	.905 ±.027	.781 ±.155	**.862 ±.041**	.567 ±.233	.607 ±.127	.587	**.745**	.893 ±.048	.406 ±.254	.793 ±.056	.398 ±.128	.579 ±.115	.489 ↓.098	.614 ↓.131	.909 ±.024	.569 ±.308	.823 ±.042	.528 ±.111	.604 ±.105	.566 ↓.021	.687 ↓.058
#2	✓	×	✓	✓	.906 ±.029	**.813 ±.159**	.840 ±.035	.553 ±.226	.604 ±.117	.578	.743	.902 ±.025	.416 ±.241	.772 ±.071	.418 ±.137	.523 ±.130	.470 ↓.108	.606 ↓.137	.911 ±.024	.606 ±.256	.832 ±.036	.545 ±.107	.595 ±.115	.570 ↓.008	.698 ↓.045
#3	×	×	✓	✓	.899 ±.044	.768 ±.195	.836 ±.059	.523 ±.235	.611 ±.137	.567	.727	.897 ±.044	.433 ±.209	.759 ±.067	.366 ±.147	.561 ±.143	.463 ↓.104	.603 ↓.124	.898 ±.031	.586 ±.292	.823 ±.039	.466 ±.148	.598 ±.093	.532 ↓.035	.675 ↓.052
#4	✓	✓	✓	×	.902 ±.033	.759 ±.218	.852 ±.049	.555 ±.229	**.638 ±.119**	.596	.741	**.908 ±.032**	**.610 ±.222**	.802 ±.090	.406 ±.128	.542 ±.145	.474 ↓.122	.654 ↓.087	**.917 ±.025**	**.734 ±.216**	.841 ±.042	.531 ±.095	.618 ±.102	.574 ↓.022	**.728 ↓.013**
#5	✓	✓	×	✓	**.910 ±.021**	.789 ±.135	.835 ±.041	.505 ±.215	.592 ±.114	.549	.726	.897 ±.024	.535 ±.191	.737 ±.099	.413 ±.137	.509 ±.138	.461 ↓.088	.618 ↓.108	.906 ±.026	.620 ±.242	.815 ±.056	.524 ±.106	.592 ±.118	.558 ↑.009	.691 ↓.035
#6	✓	✓	×	×	.903 ±.032	.777 ±.167	.830 ±.063	.523 ±.215	.585 ±.136	.554	.724	.887 ±.079	.597 ±.155	.767 ±.113	.387 ±.141	.477 ±.163	.432 ↓.122	.623 ↓.101	.912 ±.028	.700 ±.198	.832 ±.049	.529 ±.093	.589 ±.107	.559 ↑.005	.712 ↓.012
#7 (ours)	✓	✓	✓	✓	.888 ±.047	.724 ±.220	.848 ±.053	**.576 ±.236**	.631 ±.134	**.603**	.733	.877 ±.078	.543 ±.212	**.803 ±.080**	**.516 ±.109**	**.618 ±.135**	**.567 ↓.036**	**.672 ↓.061**	.915 ±.028	.659 ±.227	**.846 ±.033**	**.551 ±.101**	**.624 ±.104**	**.588 ↓.015**	.719 ↓.014

TABLE IX

PERFORMANCE COMPARISON BETWEEN OUR METHOD AND DIFFERENT VARIANTS OF BAYESEG (I.E., BAYESEG+ AND BAYESEG++). HERE, "TS": TARGETS

Method	MYOPS2020 (Source)							pA-CMR (Target)							pB-CMR (Target)								
	LV	RV	MYO	Scar	Edema	MyoPS	Avg.	LV	RV	MYO	Scar	Edema	MyoPS	Avg.	LV	RV	MYO	Scar	Edema	MyoPS	Avg.	MyoPS (Ts)	Avg. (Ts)
BayeSeg+	**.892 ±.040**	.754 ±.130	.845 ±.049	.462 ±.215	.510 ±.134	.486	.693	**.902 ±.030**	.436 ±.264	**.820 ±.062**	.295 ±.175	.440 ±.161	.368 ↓.118	.579 ↓.114	.904 ±.038	.633 ±.289	.833 ±.060	.479 ±.119	.542 ±.129	.510 ↑.024	.678 ↓.015	.439	.629
BayeSeg++	.889 ±.043	**.786 ±.174**	**.857 ±.062**	.423 ±.252	.405 ±.227	.414	.672	.881 ±.061	**.700 ±.197**	.788 ±.077	.252 ±.139	.416 ±.181	.334 ↓.080	.607 ↓.065	.906 ±.031	**.745 ±.213**	.832 ±.070	.408 ±.151	.459 ±.155	.433 ↑.019	.670 ↓.002	.384	.639
Ours	.888 ±.047	.724 ±.220	.848 ±.053	**.576 ±.236**	**.631 ±.134**	**.603**	**.733**	.877 ±.078	.543 ±.212	.803 ±.080	**.516 ±.109**	**.618 ±.135**	**.567 ↓.036**	**.672 ↓.061**	**.915 ±.028**	.659 ±.227	**.846 ±.033**	**.551 ±.101**	**.624 ±.104**	**.588 ↓.015**	**.719 ↓.014**	**.577**	**.695**

TABLE X

RESULTS OF THE EFFECTS OF CMR SEQUENCES SELECTION ON MODEL GENERALIZABILITY

Model				MYOPS2020 (Source)							pA-CMR (Target)							pB-CMR (Target)						
	$bSSFP$	LGE	$T2$	LV	RV	MYO	Scar	Edema	MyoPS	Avg.	LV	RV	MYO	Scar	Edema	MyoPS	Avg.	LV	RV	MYO	Scar	Edema	MyoPS	Avg.
#1	×	✓	✓	**.906 ±.032**	**.736 ±.236**	**.865 ±.042**	.564 ±.229	.622 ±.119	.593	**.739**	**.904 ±.041**	.484 ±.196	.797 ±.077	.415 ±.083	.590 ±.118	.503 ↓.090	.638 ↓.101	**.919 ±.022**	.649 ±.236	.838 ±.050	.519 ±.101	.581 ±.131	.550 ↓.043	.701 ↓.038
#2 (ours)	✓	✓	✓	.888 ±.047	.724 ±.220	.848 ±.053	**.576 ±.236**	**.631 ±.134**	**.603**	.733	.877 ±.078	**.543 ±.212**	**.803 ±.080**	**.516 ±.109**	**.618 ±.135**	**.567 ↓.036**	**.672 ↓.061**	.915 ±.028	**.659 ±.227**	**.846 ±.033**	**.551 ±.101**	**.624 ±.104**	**.588 ↓.015**	**.719 ↓.014**

4) Effect of CMR Sequences on Model Generalizability: To explore the effects of sequence selection on concept modeling, the performance of different sequences in modeling the pathology concept variable $\boldsymbol{o}$ was studied. As shown in Table X, Model #1 uses two sequences (i.e., LGE and T2) to model variable $\boldsymbol{o}$, while Model #2 employs three sequences (i.e., bSSFP, LGE, and T2). Compared with Model #1, Model #2 increases the Dice score of pathology segmentation from 0.503 to 0.567 on the target pA-CMR dataset and from 0.550 to 0.588 on the pB-CMR dataset. These results indicate that the bSSFP sequence may also provide pathology-related information, positively contributing to the modeling of the pathology concept. This effect may be attributed to the fact that the current bSSFP cine sequence is acquired after the injection of contrast agents.

F. Further Analyses

1) Analysis of Loss-Balancing Weights: To verify the effect of loss-balancing weights, a series of analyses were performed in Fig. 6. (a) For the variational loss weight λ_{var}, small values (e.g., 1 and 10) would limit the function of the variational loss, weakening segmentation generalizability. In contrast, a

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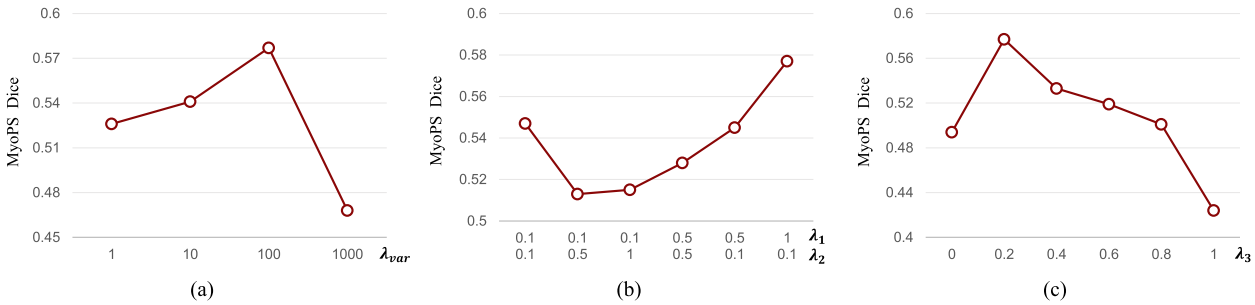


Fig. 6. Effects of loss-balancing weights on the generalizability of HSCM-Net in MyoPS. (a) Variational loss weight λ_{var}. (b) MYO-pathology constraint loss weight λ_1 and λ_2. (c) Scar-edema constraint loss weights λ_3.

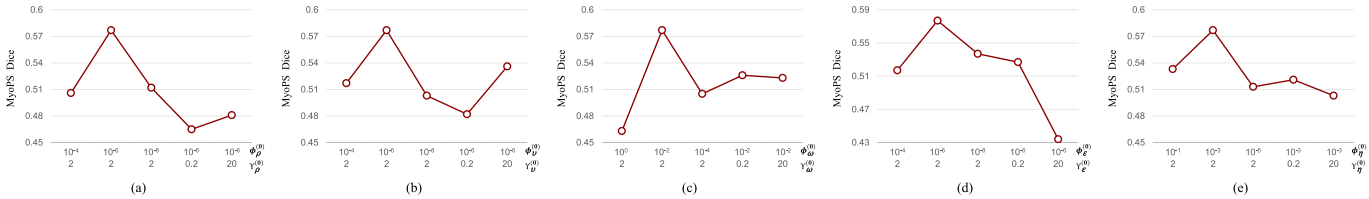


Fig. 7. Effects of prior hyperparameters on HSCM-Net for MyoPS. The different Gamma prior $\mathcal{G}(\boldsymbol{\gamma}_*^{(0)}, \boldsymbol{\phi}_*^{(0)})$ for: (a) $\boldsymbol{\rho}$, (b) $\boldsymbol{\upsilon}$, (c) $\boldsymbol{\omega}$, (d) $\boldsymbol{\varepsilon}$, and (e) $\boldsymbol{\eta}$.

TABLE XI

EVALUATION OF MODEL EFFICIENCY, INCLUDING MODEL PARAMETERS (PARAMS), TRAINING INPUT SIZE (TIS), TRAINING MEMORY (TM), TRAINING TIME (TT), INFERENCE INPUT SIZE (IIS), INFERENCE MEMORY (IM), AND INFERENCE TIME (TT)

Model	Params(M)	TIS	TM(MB)	TT(ms)	IIS	IM(MB)	IT(ms)	Dice(MyoPS)
DSU [10]	**25.76**	$8 * 3 * 196^2$	**11164**	**247**	$1 * 3 * 196^2$	**710**	78	.503
Mamba-Sea [6]	27.43	$8 * 3 * 196^2$	13972	350	$1 * 3 * 196^2$	1818	**59**	.479
Ours	27.69	$8 * 3 * 196^2$	15776	445	$1 * 3 * 196^2$	760	79	**.577**

large value (e.g., 1000) hinders the model's ability to fit training data. Therefore, $\lambda_{var} = 100$ is a suitable choice to balance data fitting and model generalizability. (b) For the MYO-pathology constraint losses λ_1 and λ_2, increasing the ratio λ_1/λ_2 consistently improves MyoPS segmentation performance. This aligns with clinical observations, as edema regions often encompass scar areas. A higher λ_1 effectively reinforces anatomical consistency between edema and myocardial structures, enhancing pathology localization. (c) For the scar-edema constraint weight λ_3, omitting it ($\lambda_3 = 0$) leads to a lack of spatial prior between pathologies, harming segmentation quality. On the other hand, an overly strong constraint ($\lambda_3 = 1$) over-regularizes the network, limiting flexibility. The best performance is observed at $\lambda_3 = 0.2$, which provides sufficient spatial guidance without compromising learning capacity.

2) Analysis of Prior Hyperparameters: To verify the effect of prior hyperparameters, a series of sensitivity analyses were performed. As shown in Fig. 7, the shape and scale parameters $\boldsymbol{\gamma}^{(0)}$ and $\boldsymbol{\phi}^{(0)}$ of the Gamma prior $\mathcal{G}(\boldsymbol{\gamma}_*^{(0)}, \boldsymbol{\phi}_*^{(0)})$ are varied for five key latent variables, i.e., $\boldsymbol{\rho}$, $\boldsymbol{\upsilon}$, $\boldsymbol{\omega}$, $\boldsymbol{\varepsilon}$, and $\boldsymbol{\eta}$. For each variable, a range of configurations is tested, spanning from strong to weak prior assumptions. Results indicate that model performance, measured by the MyoPS Dice score, is highly sensitive to these settings. For example, the best performance for $\boldsymbol{\rho}$ is achieved when $\boldsymbol{\gamma}_\rho^{(0)} = 2$ and $\boldsymbol{\phi}_\rho^{(0)} = 10^{-6}$. Both overly flat (e.g., 10^{-4}) and overly sharp (e.g., 10^{-8}) priors led to degraded performance due to poor concept modeling. This analysis demonstrates that moderately informative priors can substantially enhance generalization. Accordingly, the best-performing configurations from Fig. 7 are adopted, yielding improved segmentation performance while preserving model interpretability and probabilistic semantics.

3) Analysis of Model Efficiency: To evaluate the model efficiency of HSCM-Net, a comprehensive analysis was conducted. As shown in Table XI, HSCM-Net is compared with the second-best DG method (DSU) and the recent SOTA DG method (Mamba-Sea). During training, HSCM-Net requires 15 776 MB of memory and 445 ms per iteration, which is slightly higher than Mamba-Sea (13 972 MB, 424 ms) but remains reasonable given the model's architectural complexity. At inference, HSCM-Net demonstrates favorable efficiency, using only 760 MB of memory, which is substantially less than Mamba-Sea (1818 MB), and requiring 79 ms per image. This is comparable to DSU (78 ms) and only marginally higher than Mamba-Sea (59 ms). In terms of model size, HSCM-Net contains 27.69 M parameters, which is close to DSU (25.76 M) and Mamba-Sea (27.43 M), reflecting a compact design despite the inclusion of hierarchical semantic modeling and Bayesian inference. These results indicate that HSCM-Net achieves a balanced tradeoff between model efficiency and segmentation performance. Its inference efficiency, along

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with relatively low parameter count, suggests that it is suitable for practical deployment, even in resource-constrained clinical environments.

V. Discussion and Conclusion

This work proposes a novel framework for MyoPS, named HSCM-Net, which enhances the generalizability of the MyoPS through hierarchical semantic concept modeling of multisequence CMR images and labels. Specifically, the multi-sequence CMR images are decomposed into shape, pathology, and appearance concepts, which integrate multi-sequence information. Then, hierarchical concept priors are assigned for these three corresponding variables. Subsequently, their posterior distributions are approximated using VI, and then a DL architecture is built to implement this inference process. The experimental results on three cross-domain CMR datasets have shown the effectiveness of our proposed method. Moreover, the posterior distributions of each concept variable are further visualized, and the advantage of concept modeling in enhancing generalizability is validated.

It is worth noting that the shape and pathology concept variables are not entirely independent. Visualization in Fig. 4 shows that the shape variable ($\boldsymbol{s}$) contains a small amount of pathology information, and likewise the pathology variable ($\boldsymbol{o}$) reflects some shape-related features. Such presence of information crossover between them can impair the performance of subsequent segmentation models, particularly for the challenging pathology segmentation tasks. Therefore, separating them is crucial to ensure that variable $\boldsymbol{o}$ solely possesses pathology information. In future work, additional constraints should be applied to prevent the variable $\boldsymbol{s}$ from encapsulating pathology information, thereby improving the accuracy and generalizability. Besides, pathology was believed to be solely associated with LGE and T2 sequences, so extracting pathology concept from these two sequences appears to be more advantageous. However, the performance using all three sequences (bSSFP, LGE, and T2) actually performed better than the model using only the two sequences (LGE and T2) (see Table X). This indicates that the bSSFP sequence also contains valuable information about the pathology. In the future, it is worth exploring pathology concept modeling directly from this sequence, which holds considerable value in clinical applications.

The promising generalizability of HSCM-Net across three-domain CMR datasets demonstrates its robustness to cross-center domain shifts and its potential for clinical application. Although the training stage involves Bayesian inference with relatively high GPU memory consumption (about 15 GB, see Table XI), the inference process is lightweight and efficient, requiring only 760 MB of memory and 79 ms per image, making the method practical for routine clinical use. The marginally lower performance on the source domain compared with Mamba-Sea reflects a deliberate strategy to prevent source overfitting and prioritize domain-invariant representations. Furthermore, the slight compromise on large anatomical structures (LV, RV, and MYO) compared with BayeSeg and RandConv stems from our model prioritizing highly variable pathology concepts over stable shape priors. This tradeoff is clinically justified, as it yields greater generalizability for challenging pathology regions across unseen domains. Nevertheless, the current evaluation is limited to three-domain cardiac datasets, and further validation on larger and more diverse multicenter CMR datasets is necessary, along with exploration of the model generalizability in broader out-of-distribution scenarios, such as long-axis sequences or unknown cardiac disease types. Moreover, as the proposed hierarchical semantic concept modeling is designed for MyoPS, future research should extend this framework to other lesion types to fully verify its scalability and general applicability.

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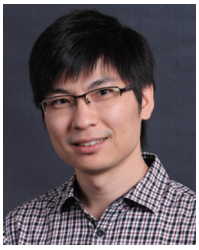
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